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COMPUTING DEVICE CASE WITH REMOVABLE KICKSTAND

Abstract

A case for a computing device comprises a removable kickstand. In one example, a case substrate comprises a friction clip mount. A kickstand is removably and rotatably coupled to the case substrate via a friction clip that partially surrounds the friction clip mount. The friction clip defines an opening through which the friction clip mount is removable from the friction clip to thereby separate the kickstand from the case substrate.

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Background/Summary

BACKGROUND

[0001] Some computing devices, such as a tablet computing device, utilize a case that includes a kickstand. The kickstand is movable to support at least a portion of the device, such as by propping up a display portion of the tablet computing device on a tabletop.

SUMMARY

[0002] According to one aspect of the present disclosure, a case is provided for a computing device. The case comprises a case substrate that includes a friction clip mount. A removable kickstand includes a friction clip that is removably coupled to the friction clip mount. The friction clip defines an opening that partially surrounds through the friction clip mount to rotatably couple the kickstand to the case substrate. The friction clip is removeable from the friction clip mount via the opening to thereby separate the kickstand from the case substrate.

[0003] According to another aspect of the present disclosure, a case for a computing device comprises a removable kickstand comprising a friction clip mount. A case substrate comprises a friction clip that is removably coupled to the friction clip mount, with the friction clip defining an opening that partially surrounds the friction clip mount to rotatably couple the kickstand to the case substrate. The friction clip is removeable from the friction clip mount via the opening to thereby separate the kickstand from the case substrate. The case substrate further comprises a surface relief feature operatively configured to disengage the friction clip from the friction clip mount when the kickstand is rotated to an angle greater than a disengagement threshold angle between the kickstand and the case substrate.

[0004] In another aspect of the present disclosure, a method for detaching a kickstand from a case for a computing device is disclosed. The kickstand is removably and rotatably coupled to a case substrate of the case via a friction clip that partially surrounds a friction clip mount of the case substrate. The method includes rotating the kickstand to form a rotated angle between the kickstand and the case substrate of greater than a disengagement threshold angle and, at least at the rotated angle, leveraging the kickstand against at least a portion of the case substrate to force the friction clip mount to separate from the friction clip through an opening in the friction clip.

[0005] This Summary is provided to introduce a selection of concepts in a simplified form that are further described below in the Detailed Description. This Summary is not intended to identify key features or essential features of the claimed subject matter, nor is it intended to be used to limit the scope of the claimed subject matter. Furthermore, the claimed subject matter is not limited to implementations that solve any or all disadvantages noted in any part of this disclosure.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0006] FIG. **1** shows an example of a case for a computing device according to an example embodiment of the subject disclosure.

[0007] FIG. **2** shows an exploded view of the case of FIG. **1** with the kickstand detached from the case substrate.

[0008] FIG. **3** shows a cross-sectional view of a portion of the case of FIG. **1**.

[0009] FIG. **4** shows the same cross-sectional view with the kickstand of the case in a closed configuration.

[0010] FIG. **5** shows the same cross-sectional view with the kickstand rotated to an angle greater than a disengagement threshold angle.

[0011] FIG. **6** shows an enlarged view of a friction clip and a friction clip mount that can be used in the case of FIG. **1**.

[0012] FIG. **7** shows another view of the case of FIG. **1**.

[0013] FIG. **8** shows another cross-sectional view of the portion of the case depicted in FIGS. **3-5** with the kickstand being rotated towards the closed configuration.

[0014] FIG. **9** shows another example of a case for a computing device according to an example of the present disclosure.

[0015] FIG. **10** shows an exploded view of the case of FIG. **9** with the kickstand detached from the case substrate.

[0016] FIG. **11** shows a cross-sectional view of a portion of the case of FIG. **9**.

[0017] FIG. **12** shows the same cross-sectional view with the kickstand rotated to an open configuration.

[0018] FIG. **13** shows the same cross-sectional view with the kickstand rotated to an angle greater than a disengagement threshold angle.

[0019] FIG. **14** is a flowchart illustrating a method for detaching a kickstand removably and rotatably coupled to a case substrate for a computing device via a friction clip that partially surrounds a friction clip mount of the case substrate according to an example embodiment of the subject disclosure.

DETAILED DESCRIPTION

[0020] As introduced above, some computing devices utilize a case that includes a kickstand. The kickstand is operatively configured to support at least a portion of the device. For example, some tablet computing devices include a kickstand that rotates to an angled position with respect to the device's display and allows the device to stand at an angled position on a surface, such as a tabletop. However, because kickstands are rotatable and protrude from a housing of the computing device, they are subject to potential misuse, unintended over-rotations by inattentive users, and accidental drops or other impacts. These potentially damaging use cases are more common in rugged use environments, such as outdoors and in industrial environments, and with certain groups of users such as children and persons with limited dexterity. In these cases, damage to the kickstand and/or one or more hinges coupling the kickstand to the device can result.

[0021] To address one or more of these issues, a case that includes a rotatable and removable kickstand is provided for a computing device. The case includes a case substrate comprising a friction clip mount. The kickstand comprises friction clip that is removably coupled to a friction clip mount. The friction clip defines an opening that partially surrounds the friction clip mount to rotatably couple the kickstand to the case substrate. When connected, the friction clip cooperates with the friction clip mount to provide sufficient friction to enable the kickstand to support at least a portion of the computing device at different angles. Advantageously and as described in more detail below, friction clip is removeable from the friction clip mount via the opening to thereby separate the kickstand from the case substrate. More particularly, the friction clip is configured to disengage from the friction clip mount in response to rotation of the kickstand beyond a disengagement threshold angle, thereby allowing the kickstand to safely detach from the case substrate without damaging the kickstand or the substrate.

[0022] FIG. **1** shows a rear perspective view of one example of a case **100** for a computing device, such as a laptop computing device, a tablet computing device, or a mobile computing device (e.g., a smartphone), according to examples of the present disclosure. It will also be appreciated that the case **100** may be configured for use with any other suitable computing device. Other examples of suitable computing devices include, but are not limited to, a home-entertainment computing device, a network computing device, and a gaming device.

[0023] In some examples, the case **100** is an integral chassis of the computing device that houses internal components of the device. In other examples, the case **100** is a removable external case that at least partially surrounds a chassis of the computing device.

[0024] The case **100** comprises a case substrate **102** that includes an external surface in the form of rear surface **126**. In the present example, the case **100** is configured to contain a tablet computing device (not shown) such that the display of the device faces away from rear surface **126**. The case **100** and case substrate **102** are configured to protect an enclosed computing device from impacts, scratches, drops, and other forces that could potentially damage the device. In different examples, the case **100** is fabricated from a thermoplastic polymer, such as polycarbonate or acrylonitrile butadiene styrene (ABS), one or more elastomers such as silicone rubbers, combinations of the foregoing, and/or any other suitable materials.

[0025] The case **100** also includes a kickstand **104** that is rotatably coupled to the case substrate **102**, as described further below. The kickstand **104** is operatively configured to support at least a portion of the device, such as by propping up the device by placing the rear edge **105** of the kickstand and a bottom edge **107** of the case **100** on a surface such as a tabletop. This may enable a user to interact with the device more easily as compared to the device laid flat on the surface. For example, leaning a tablet computing device against a kickstand may allow users to view displayed content from a distance without holding the device.

[0026] As shown in FIG. **2**, and in one potential advantage of the present disclosure, the kickstand **104** is also removable from the case substrate **102**. For example, as introduced above and described in more detail below, the kickstand **104** is configured to separate from the case substrate **102** in response to rotation of the kickstand beyond a disengagement threshold angle to prevent damage to the kickstand or the case substrate.

[0027] In the present example and as described further below, the kickstand **104** is removably and rotatably coupled to the case substrate **102** via a plurality of friction clips **106**, **154**, **166**, **168**, **170**, and **172** and corresponding friction clip mounts **108**, **156**, **174**, **176**, **178** and **180**. In other examples, cases of the present disclosure can utilize a single friction clip and corresponding friction clip mount, or any number of friction clips and friction clip mounts. In the present example, the following description of friction clip **106** and corresponding friction clip mount **108** is applicable to the other friction clips **154**, **166**, **168**, **170**, and **172** and corresponding friction clip mounts **156**, **174**, **176**, **178** and **180**.

[0028] FIG. **3** shows a cross-sectional view of a portion of the case **100** along line **3-3** of FIG. **1**. As shown in FIG. **3**, in this example the friction clip **106** comprises a partial cylindrical tube. In other examples, the friction clip **106** may comprise any other suitable shape. Other examples of suitable shapes include, but are not limited to, a ball socket and other shapes complementary to a shape of a clip mount to which the friction clip is attached.

[0029] In this example, the friction clip **106** partially surrounds a friction clip mount **108** of the case substrate **102** in the form of an elongated cylindrical pin. In the present example, the cylindrical pin extends laterally across the case substrate **102** to form the other friction clip mounts **156**, **174**, **176**, **178** and **180**. In other examples, the other friction clip mounts **156**, **174**, **176**, **178** and **180** can be formed from separate pins. In other examples, a friction clip mount may comprise any other suitable shape complementary to a shape of a friction clip to which the friction clip mount is attached. Other examples of suitable shapes include, but are not limited to, spheroids and other shapes complementary to a shape of the friction clip.

[0030] In the present example and with reference also to FIG. **2**, the case substrate **102** comprises the friction clip mounts **108**, **156**, **174**, **176**, **178** and **180**, and the kickstand **104** comprises the friction clips **106**, **154**, **166**, **168**, **170**, and **172**. In other examples, and as described in more detail below with reference to FIGS. **9-13**, the case substrate comprises one or more friction clips and the kickstand comprises corresponding friction clip mount(s).

[0031] With reference to FIGS. **2**, **3**, and **5**, in this example the friction clip **106** comprises a partial cylindrical tube that defines an opening **110** through which the friction clip mount **108** is attachable to and removable from the friction clip. In this manner, in one potential advantage of the present disclosure and as described further below, when the kickstand **104** is attached to the case substrate **102** via the coupling of the friction clip **106** and the friction clip mount **108**, the opening **110** enables the kickstand **104** to be separated from the case substrate **102**, thereby reducing the risk of damage to the kickstand and/or case due to over-rotation of the kickstand.

[0032] In the example of case **100** and as depicted in FIG. **2**, the opening **110** comprises a slot **112** that extends along a length of the partial cylindrical tube comprising the friction clip **106**. The friction clip mount **108** resides in the slot **112** when the kickstand **104** is coupled to the case substrate **102**. In this manner, the friction clip **106** and the friction clip mount **108** form a hinge having an axis of rotation located at a center point of and extending parallel to the cylindrical pin/friction clip mount. In this manner, the kickstand **104** is rotatable with respect to the case substrate **102** around the longitudinal axis of the friction clip mount **108**.

[0033] More particularly, as schematically illustrated in the examples of FIGS. **3-4** and described in more detail below, in this example the hinge formed by the friction clip **106** and the friction clip mount **108** cooperates with a leading edge **144** of the case substrate **102** to permit the kickstand **104** to rotate to a disengagement threshold angle **124** of approximately 75 degrees between a closed configuration (FIG. **4**) and an open configuration (FIG. **3**). Additionally, in another potential advantage of the present disclosure and as described in more detail below with reference to FIG. **5**, rotation of the kickstand **104** to an angle greater than the disengagement threshold angle **124** causes the kickstand **104** to separate from the case substrate **102**. Advantageously, this prevents damage to the kickstand **104** and/or the case substrate **102** that could occur due to overextension of the kickstand. In other examples and configurations of cases according to the present disclosure, other disengagement threshold angles can be utilized. For example, the kickstand **104** can be rotatable relative to the rear surface **126** of the case substrate **102** to a disengagement threshold angle of up to 45 degrees, up to 60 degrees, up to 90 degrees, up to 120 degrees, or other angle of less than 180 degrees.

[0034] With continued reference to FIG. **3**, the kickstand **104** is operatively configured to be coupled to the case substrate **102** by snapping the friction clip **106** onto the friction clip mount **108**. In some examples, the friction clip **106** comprises beveled edges **114**, **116** adjacent to the opening **110**. In another potential advantage of the present disclosure, the beveled edges **114**, **116** are configured to guide the friction clip **106** over the friction clip mount **108** through the opening **110** and into slot **112**, thereby enabling a user to easily align and snap the friction clip onto the elongated pin of the friction clip mount to attach the kickstand **104** to the case substrate **102**.

[0035] As depicted in FIG. **6**, which shows an exploded view of the friction clip **106** and the friction clip mount **108**, the slot **112** has an internal diameter **118** that is less than a diameter **120** of the cylindrical pin of friction clip mount **108** when the pin is not located within the slot. Additionally, the friction clip **106** is fabricated a material having some measure of flexibility that enables the friction clip to expand to receive the friction clip mount **108** and form a rotatable engagement embodying a frictional resistance to rotation. Some examples of suitable materials for friction clip **106** include, but are not limited to, acrylonitrile butadiene styrene (ABS), polycarbonate, spring steel, and glass-filled polymers. In this manner, the friction clip **106** can undergo elastic deformation that allows the friction clip mount **108** to pass through the opening **110** and be retained within the slot **112**.

[0036] As the friction clip **106** is undersized relative to the diameter of the friction clip mount **108**, the friction clip **106** exerts a force on the friction clip mount **108** that resists rotation of the kickstand **104** relative to the case substrate **102**. In different examples, the internal diameter **118** of the slot **112** and the external diameter **120** of the friction clip mount **108** are configured to provide a magnitude of frictional engagement that enables the kickstand **104** to remain in an angular orientation set by a user, and correspondingly resist rotation under the weight of the computing device housed within the case substrate **102**, while also allowing the user to reposition the kickstand by applying a relatively small magnitude of torque, such as approximately 2.0 Newton meters (Nm). Accordingly, the internal diameter **118** of the slot **112** and the external diameter **120** of the friction clip mount are configurable to provide the case with a suitably stiff or loose hinge for

an end-user application.
[0037] In some examples, a supplemental frictional component is utilized in the slot **112**. In the example of FIG. **6**, a friction band **122** is molded into the friction clip **106**. The friction band **122** at least partially covers an internal surface of the slot **112**, such that the friction band **122** is in contact with the friction clip mount **108** when the friction clip mount is seated within the friction clip. The friction band **122** is configured to generate additional friction and resistance to rotation between the friction clip **106** and the friction clip mount **108**. Accordingly, and in one potential advantage of this example, the friction band **122** enables the setting of a desired frictional resistance to rotation in a structurally efficient manner, as the friction band occupies a relatively small additional structural volume.
[0038] The friction band **122** optionally comprises a different material than the friction clip **106**. Some examples of suitable materials for the friction band **122** include, but are not limited to, rubber and other polymers having a suitably high coefficient of friction to resist rotation about the friction clip mount **108**. In other examples, the friction band **122** comprises the same material as the friction clip **106**.
[0039] In some examples, the friction band **122** and the friction clip **106** are molded together as a unitary part. In other examples, the friction band **122** and the friction clip **106** comprise separate parts that are attached together while fabricating the friction clip **106**. In this manner, the friction band **122** further reduces the internal diameter **118** of at least a portion of the friction clip **106** and/or provides additional friction and resistance to rotation between the friction clip and the friction clip mount.
[0040] As introduced above with reference to FIGS. **3-5**, the kickstand **104** is rotatable through a range of angular orientations with respect to the case substrate **102**. In some examples, the kickstand **104** is flush with the rear surface **126** of the case substrate **102** when an angle between the kickstand **104** and the case substrate **102** is zero degrees (e.g., the kickstand is in a closed configuration), as depicted in the example of FIG. **4**. In such examples, the kickstand **104** is configured to reside in a recessed portion **128** of the case substrate **102** adjacent to a recessed surface **129** when the kickstand is in the closed configuration (See also FIGS. **2** and **3**). In this manner, the kickstand **104** can be completely stowed within the case substrate **102** such that it does not protrude past the rear surface **126**. Accordingly, in other potential advantages of the present disclosure, in this stowed orientation the kickstand **104** is protected from damage that could occur from impacts to protruding parts, the case **100** is easy and comfortable for a user to handle, store, and transport, and unintentional deployment of the kickstand (e.g., by snagging the kickstand on an object) is prevented.
[0041] With reference to FIGS. **2-4** and **7**, the kickstand **104** includes a friction clip support structure **130** that extends from the friction clip **106**. The case substrate **102** also includes an aperture **132** defined in the recessed surface **129** and configured to receive the friction clip support structure **130** when the kickstand **104** is rotated into the closed configuration of FIG. **4**. FIG. **7** shows another view of the case **100** illustrating the spatial relationship between the friction clip support structure **130** and the aperture **132**. FIG. **7** depicts the kickstand **104** attached to the case substrate **102** in an open configuration. In the closed configuration depicted in the example of FIG. **4**, at least a portion of the friction clip support structure **130** is located within the aperture **132**. This enables the kickstand to be stowed within the case substrate **102** as described above.
[0042] With reference now to FIG. **8**, in this example a portion of the recessed surface **129** of the case substrate **102** forms a ramp **134**. The ramp **134** is operatively configured to contact a distal end **136** of the friction clip support structure **130** opposite the friction clip **106** when a user is reattaching the kickstand **104** and friction clip to the friction clip mount **108**. More particularly, when a user inserts the friction clip support structure **130** and friction clip **106** within the aperture **132**, the ramp **134** is operatively configured to urge the friction clip **106** toward the friction clip mount **108** when the kickstand **104** is rotated to an angle less than an engagement threshold angle

138 with respect to the case substrate **102**. As the kickstand **104** is rotated in the direction of arrow **150** towards the closed configuration, the distal end **136** contacts the ramp **134** which moves the friction clip **106** in the direction of arrow **152** to snap the friction clip onto the friction clip mount **108**. Accordingly, and in another potential advantage of the present disclosure, this configuration enables a user to easily attach the kickstand **104** to the case substrate **102** and ensures that the friction clip **106** engages with the friction clip mount **108** each time the kickstand is attached.

[0043] In this example the distal end **136** of the friction clip support structure **130** defines a detent **140**. The detent **140** is complementary to a shape of the ramp **134**. In this manner, the detent **140** is operatively configured to cooperate with the ramp **134** to secure the kickstand **104** in the closed configuration, as depicted in the example of FIG. **4**. Accordingly, and in another potential advantage of the present disclosure, utilizing the detent **140** and corresponding ramp **134** enables the kickstand **104** to be securely retained in the closed configuration without using magnets, clips, or other additional attachment mechanisms.

[0044] As shown in FIGS. **3-5** and **8**, in some examples the case substrate **102** defines a recessed concavity **142** adjacent to the aperture **132**. The recessed concavity **142** is opposite the ramp **134** and extends under the rear surface **126** of the case substrate **102**. The friction clip mount **108** is located within the recessed concavity **142**. In this manner, the concavity **142** is configured to accommodate the friction clip **106** when the kickstand **104** is coupled to the case substrate **102**. Accordingly, and in another potential advantage of the present disclosure, by locating the friction clip mount **108** within the recessed concavity **142**, configurations of the present disclosure protect the friction clip mount and attached friction clip **106** from external contact and/or foreign material or debris, along with hiding at least portions of the friction clip mount and attached friction clip from view.

[0045] With continued reference to FIGS. **3-5** and **8**, the case substrate **102** further comprises a leading edge **144** adjacent to the recessed concavity **142**. As shown in FIG. **3**, a neck **111** of the kickstand **104** is configured to contact the leading edge **144** when the kickstand **104** is rotated to the disengagement threshold angle **124**. In this manner, the configuration and placement of the leading edge **144** may define the angle beyond which rotation of the kickstand **104** causes the friction clip **106** to disengage from the friction clip mount **108**. Additionally, and in another potential advantage of the present disclosure, the leading edge **144** further protects the friction clip mount and attached friction clip **106** from external contact and/or foreign material or debris, along with hiding at least portions of the friction clip mount and attached friction clip from view.

[0046] Further, the leading edge **144** is operatively configured to disengage the friction clip from the friction clip mount when the kickstand is rotated to an angle greater than the disengagement threshold angle **124** with respect to the case substrate. For example, and as depicted in FIG. **5**, the neck **111** of kickstand **104** is leveraged against the leading edge **144** such that applying additional force to continue to rotate the kickstand **104** away from the case substrate **102**, as indicated at **146**, urges the friction clip **106** away from the friction clip mount **108**, as indicated at **148**. This causes the friction clip **106** to detach from the friction clip mount **108**, thus separating the kickstand **104** from the case substrate **102**.

[0047] As depicted in FIG. **3**, the opening **110** in the friction clip **106** faces into the recessed concavity **142** when the kickstand is oriented at the disengagement threshold angle **124**. In this manner, the friction clip **106** can disengage from the friction clip mount **108** when urged away from the friction clip mount by the force created by the leading edge **144** in the y-axis direction. In this manner, the kickstand **104** is configured to separate from the case substrate **102** in response to pressuring the kickstand **104** to rotate past the disengagement threshold angle **124**.

[0048] In some examples and as noted above, the kickstand **104** comprises two or more friction clips and two or more corresponding friction clip mounts. In some configurations, such as the example shown in FIGS. **1-8**, utilizing multiple friction clips and friction clip mounts enables the kickstand **104** to be mounted to the case substrate **102** with additional strength and stability as compared to the use of a single friction clip and friction clip mount. In addition, providing a plurality of smaller friction clips can enable a user to connect the kickstand **104** to the case substrate **102** more easily as compared to providing a single, larger friction clip.

[0049] With reference now to FIG. **2**, in this example each friction clip mount **108**, **174**, **178** is separated from an adjacent friction clip mount **156**, **176**, **180**, respectively, by a friction clip mount support **158**. Correspondingly, on the kickstand **104** each friction clip **106**, **166**, **170** is separated from an adjacent friction clip **154**, **168**, **172**, respectively, by a gap **160**. The gaps **160** are configured to accommodate the friction clip mount supports **158** when the friction clips **106**, **154**, **166**, **168**, **170**, and **172** are rotatably attached to their corresponding friction clip mounts **108**, **156**, **174**, **176**, **178** and **180**. In another potential advantage of the present disclosure, the friction clip mount supports **158** enhance the strength and rigidity of the friction clip mounts that extend through the supports. For example, where the friction clip mounts **108**, **156**, **174**, **176**, **178** and **180** are a single elongated cylindrical pin, the friction clip mount supports **158** provide structural integrity to the pin to facilitate durability and extended duty cycles of attaching and detaching the kickstand. The friction clip mount supports **158** also can cooperate with the gaps **160** to assist a user in attaching the kickstand **104** by guiding the friction clips **106**, **154**, **166**, **168**, **170**, and **172** onto their corresponding friction clip mounts **108**, **156**, **174**, **176**, **178** and **180**.

[0050] In the examples described above with reference to FIGS. **1-8**, the friction clips are located at the kickstand **104** and the friction clip mounts are located at the case substrate **102**. FIGS. **9** and **10** show another example of a case **900** for a computing device according to examples of the present disclosure. Like the case **100** of FIGS. **1-8**, the case **900** comprises a case substrate **902** and a kickstand **904**. In contrast to the case **100** of FIGS. **1-8**, the kickstand **904** comprises friction clip mounts **908** and the case substrate **902** comprises friction clips **906** that are configured to partially surround the friction clip mount **908** to removably and rotatably couple the kickstand **904** to the case substrate **902**.

[0051] With reference also to FIG. **11**, and like the friction clip **106** of the example case **100** depicted in FIGS. **1-8**, the friction clips **906** each comprise a partial cylindrical tube. In other examples, the friction clips **906** may comprise any other suitable shape. Other examples of suitable shapes include, but are not limited to, a ball socket and other shapes complementary to a shape of a friction clip mount to which the friction clip is attached.

[0052] Like the friction clip mount **108** of the example case **100** depicted in FIGS. **1-8**, the friction clip mounts **908** each comprise an elongated cylindrical pin. In the present example, a single cylindrical pin extends laterally across a front edge **920** of the kickstand **904** to form the friction clip mounts **908**. In other examples, each of the friction clip mounts **908** is formed from separate cylindrical pins. In other examples, the friction clip mounts **908** may comprise any other suitable shape. Other examples of suitable shapes include, but are not limited to, spheroids and other shapes complementary to a shape of the friction clips **906**.

[0053] FIG. **11** shows a cross-sectional view of a portion of the case **900** along line **11-11** of FIG. **9**. As shown in FIG. **11**, the friction clip **906** defines an opening **910** that partially surrounds the friction clip mount **908** to rotatably couple the kickstand **904** to the case substrate **902**. In this manner, the friction clip mount **908** is removeable from the friction clip **906** via the opening **910** to thereby separate the kickstand **904** from the case substrate **902**. Like the openings **110** in the friction clips of the example case **100** depicted in FIGS. **1-8**, each of the openings **910** in friction clips **906** comprises a slot that extends along a length of the partial cylindrical tube of the friction clip.

[0054] As illustrated by example in FIGS. **12** and **13**, and like the kickstand **104** of FIGS. **1-8**, the kickstand **904** is configured to separate from the case substrate **902** upon rotation of the kickstand **904** beyond a disengagement threshold angle **912** between the kickstand **904** and the case substrate **902**. This prevents damage to the kickstand or the case substrate that could occur due to overextension of the kickstand.

[0055] The case substrate **902** further comprises a surface relief feature **914**. The surface relief feature **914** is operatively configured to disengage the friction clip **906** from the friction clip mount **908** when the kickstand **904** is rotated to an angle greater than the disengagement threshold angle **912**. In the example depicted in FIGS. **9-13**, the surface relief feature **914** extends laterally in the x-axis direction across a width of a rear surface **926** of the case substrate **902**. With reference to FIGS. **11** and **12**, the surface relief feature **914** abuts a trough **930** formed by the surface relief feature and the friction clip **906**. The surface relief feature **914** is positioned proud of the trough **930**, and the trough provides space for the kickstand **904** to rotate to the disengagement threshold angle **912**. In this manner, the surface relief feature **914** is operatively configured to contact the kickstand **904** when the kickstand is rotated to and beyond the disengagement threshold angle **912**. In this manner and in different configurations, the placement of the surface relief feature **914** can be utilized to define the disengagement threshold angle beyond which rotation of the kickstand **904** causes the friction clip **906** to disengage from the friction clip mount **908**.
[0056] In the example of FIGS. **12** and **13**, the kickstand **904** is leveraged against the surface relief feature **914** in a manner similar to the kickstand **104** and the leading edge **144** of FIG. **5**. In this manner, applying additional force to continue to rotate the kickstand **904** beyond the disengagement threshold angle **912**, as indicated at **916**, urges the friction clip mount **908** away from the friction clip **906**, as indicated at **918**. This causes the friction clip mount **908** to detach from the friction clip **906**, thus separating the kickstand **904** from the case substrate **902**.
[0057] In this configuration, the opening **910** in the friction clip **906** faces away from the surface relief feature **914** of the case substrate **102**. In this manner, the friction clip mount **908** can disengage from the friction clip **906** when urged away from the friction clip by the force leveraged against the surface relief feature **914**. In this manner, the kickstand **904** is configured to separate from the case substrate **902** in response to rotating the kickstand past the disengagement threshold angle.
[0058] With reference now to FIG. **14**, a method **1400** is provided for detaching a kickstand that is removably and rotatably coupled to a case substrate for a computing device via a friction clip that partially surrounds a friction clip mount of the case substrate. The following description of the method **1400** is provided with reference to the components described above. For example, the method **1400** may be performed using the case **100** of FIGS. **1-8**. It will also be appreciated that the method **1400** also may be performed in other contexts using other suitable components.
[0059] At **1402**, the method **1400** comprises rotating the kickstand to form a rotated angle between the kickstand and the case substrate of greater than a disengagement threshold angle. For example, the kickstand **104** of FIGS. **3** and **5** is rotated to a rotated angle greater than the disengagement threshold angle **124**.
[0060] The method **1400** further comprises, at **1404**, at least at the rotated angle, leveraging the kickstand against at least a portion of the case substrate to force the friction clip mount to separate from the friction clip through an opening in the friction clip. For example and as shown in FIG. **5**, as the kickstand is rotated past the disengagement threshold angle **124** to the rotated angle, the kickstand **104** is leveraged against the leading edge **144** of the case substrate **102** to cause the friction clip **106** to detach from the friction clip mount **108**, thus separating the kickstand **104** from the case substrate **102**.
[0061] The above-described apparatuses and methods may be used to enable a kickstand to rotate and to support the weight of at least a portion of a computing device. The kickstand is also removable from the case and is configured to disengage in response to being rotated beyond a disengagement threshold angle. This enables the kickstand to separate from the case and prevent damage to the kickstand or case substrate.
[0062] The following paragraphs provide additional support for the claims of the subject application. One aspect provides a case for a computing device, the case comprising: a case substrate comprising a friction clip mount; and a removeable kickstand comprising a friction clip

removably coupled to the friction clip mount, wherein the friction clip defines an opening that partially surrounds the friction clip mount to rotatably couple the kickstand to the case substrate, and the friction clip is removeable from the friction clip mount via the opening to thereby separate the kickstand from the case substrate. In some examples, the friction clip mount additionally or alternatively comprises an elongated cylindrical pin. In some examples, the case substrate additionally or alternatively comprises an external surface, and the case substrate is configured such that the kickstand is flush with the external surface when an angle between the kickstand and the case substrate is zero degrees. In some examples, the kickstand additionally or alternatively comprises a friction clip support structure extending from the friction clip, wherein at least a portion of the friction clip support structure is located within an aperture defined in a recessed surface of the case substrate when the kickstand is in a closed configuration. In some examples, the case substrate additionally or alternatively defines a recessed concavity adjacent to the aperture, and the friction clip mount is located within the recessed concavity. In some examples, the case substrate additionally or alternatively comprises a leading edge adjacent to the recessed concavity, the leading edge operatively configured to contact the kickstand and disengage the friction clip from the friction clip mount when the kickstand is rotated to an angle greater than a disengagement threshold angle with respect to the case substrate. In some examples, a neck of the kickstand is configured to contact the leading edge when the kickstand is rotated to the disengagement threshold angle. In some examples, the opening in the friction clip additionally or alternatively faces into the recessed concavity when the kickstand is oriented at the disengagement threshold angle. In some examples, a portion of the recessed surface of the case substrate additionally or alternatively forms a ramp within the aperture, the ramp operatively configured to contact a distal end of the friction clip support structure and urge the friction clip toward the friction clip mount when the kickstand is rotated to an angle less than an engagement threshold angle with respect to the case substrate. In some examples, the distal end of the friction clip support structure additionally or alternatively defines a detent operatively configured to cooperate with the ramp to secure the kickstand in the closed configuration. In some examples, the friction clip additionally or alternatively comprises a partial cylindrical tube that defines the opening. In some examples, the tube additionally or alternatively defines a slot-like opening that extends along a length of the partial cylindrical tube. In some examples, the case additionally or alternatively comprises a friction band molded into the friction clip. In some examples, the kickstand additionally or alternatively comprises two or more friction clips, and wherein each friction clip of the two or more friction clips is separated from an adjacent friction clip by a gap. In some examples, the case substrate additionally or alternatively comprises two or more friction clip mounts, and wherein each friction clip mount of the two or more friction clip mounts is separated from an adjacent friction clip mount by a friction clip mount support.

[0063] Another aspect provides a case for a computing device, the case comprising: a removable kickstand comprising a friction clip mount; and a case substrate comprising: a friction clip removably coupled to the friction clip mount of the removable kickstand, the friction clip defining an opening that partially surrounds the friction clip mount to rotatably couple the kickstand to the case substrate, the friction clip mount being removeable from the friction clip via the opening to thereby separate the kickstand from the case substrate; and a surface relief feature operatively configured to disengage the friction clip from the friction clip mount when the kickstand is rotated to an angle greater than a disengagement threshold angle between the kickstand and the case substrate. In some examples, the surface relief feature comprises a projection that protrudes from an end of the rear surface of the case substrate. In some examples, the surface relief feature is positioned above a trough that is located between the surface relief feature and the friction clip. In some examples, the friction clip mount additionally or alternatively comprises an elongated cylindrical pin. In some examples, the friction clip additionally or alternatively comprises a partial cylindrical tube that defines the opening. In some examples, the tube additionally or alternatively

defines a slot-like opening that extends along a length of the partial cylindrical tube. In some examples, the opening in the friction clip additionally or alternatively faces away from the surface relief feature of the case substrate.
[0064] Another aspect provides a method for detaching a kickstand removably and rotatably coupled to a case substrate for a computing device via a friction clip that partially surrounds a friction clip mount of the case substrate, the method comprising: rotating the kickstand to form a rotated angle between the kickstand and the case substrate of greater than a disengagement threshold angle; and at least at the rotated angle, leveraging the kickstand against at least a portion of the case substrate to force the friction clip mount to separate from the friction clip through an opening in the friction clip.
[0065] It will be understood that the configurations and/or approaches described herein are exemplary in nature, and that these specific embodiments or examples are not to be considered in a limiting sense, because numerous variations are possible. The specific routines or methods described herein may represent one or more of any number of processing strategies. As such, various acts illustrated and/or described may be performed in the sequence illustrated and/or described, in other sequences, in parallel, or omitted. Likewise, the order of the above-described processes may be changed.
[0066] The subject matter of the present disclosure includes all novel and non-obvious combinations and sub-combinations of the various processes, systems and configurations, and other features, functions, acts, and/or properties disclosed herein, as well as any and all equivalents thereof.

Claims

1. A case for a computing device, the case comprising: a case substrate comprising a friction clip mount located within a recessed concavity, and a leading edge provided adjacent to the recessed concavity; and a removeable kickstand comprising a friction clip removably coupled to the friction clip mount, wherein the friction clip defines an opening that partially surrounds the friction clip mount to rotatably couple the kickstand to the case substrate, and the friction clip is removeable from the friction clip mount via the opening to thereby separate the kickstand from the case substrate; wherein the leading edge is operatively configured to contact the kickstand and disengage the friction clip from the friction clip mount when the kickstand is rotated to an angle greater than a disengagement threshold angle with respect to the case substrate.
2. The case of claim 1, wherein the friction clip mount comprises an elongated cylindrical pin.
3. The case of claim 1, wherein the case substrate comprises an external surface, and the case substrate is configured such that the kickstand is flush with the external surface when an angle between the kickstand and the case substrate is zero degrees.
4. The case of claim 1, wherein the kickstand further comprises a friction clip support structure extending from the friction clip, wherein at least a portion of the friction clip support structure is located within an aperture defined in a recessed surface of the case substrate when the kickstand is in a closed configuration.
5. The case of claim 4, wherein the recessed concavity is adjacent to the aperture.
6. The case of claim 1, wherein the opening in the friction clip faces into the recessed concavity when the kickstand is oriented at the disengagement threshold angle.
7. The case of claim 1, wherein a portion of the recessed surface of the case substrate forms a ramp within the aperture, the ramp operatively configured to contact a distal end of the friction clip support structure and urge the friction clip toward the friction clip mount when the kickstand is rotated to an angle less than an engagement threshold angle with respect to the case substrate.
8. The case of claim 7, wherein the distal end of the friction clip support structure defines a detent operatively configured to cooperate with the ramp to secure the kickstand in the closed

configuration.
9. The case of claim 1, wherein the friction clip comprises a partial cylindrical tube that defines the opening.
10. The case of claim 9, wherein the tube defines a slot-like opening that extends along a length of the partial cylindrical tube.
11. The case of claim 1, further comprising a friction band molded into the friction clip.
12. The case of claim 1, wherein the kickstand comprises two or more friction clips, and wherein each friction clip of the two or more friction clips is separated from an adjacent friction clip by a gap.
13. The case of claim 1, wherein the case substrate comprises two or more friction clip mounts, and wherein each friction clip mount of the two or more friction clip mounts is separated from an adjacent friction clip mount by a friction clip mount support.
14. A case for a computing device, the case comprising: a removable kickstand comprising a friction clip mount; and a case substrate comprising: a friction clip removably coupled to the friction clip mount of the removable kickstand, the friction clip defining an opening that partially surrounds the friction clip mount to rotatably couple the kickstand to the case substrate, the friction clip mount being removeable from the friction clip via the opening to thereby separate the kickstand from the case substrate; and a surface relief feature operatively configured to disengage the friction clip from the friction clip mount when the kickstand is rotated to an angle greater than a disengagement threshold angle between the kickstand and the case substrate.
15. The case of claim 14, wherein the friction clip mount comprises an elongated cylindrical pin.
16. The case of claim 14, wherein the friction clip comprises a partial cylindrical tube that defines the opening.
17. The case of claim 16, wherein the tube defines a slot-like opening that extends along a length of the partial cylindrical tube.
18. The case of claim 14, wherein the opening in the friction clip faces away from the surface relief feature of the case substrate.
19. A method for detaching a kickstand removably and rotatably coupled to a case substrate for a computing device via a friction clip that partially surrounds a friction clip mount of the case substrate, the method comprising: rotating the kickstand to form a rotated angle between the kickstand and the case substrate of greater than a disengagement threshold angle; and at least at the rotated angle, leveraging the kickstand against at least a portion of the case substrate to force the friction clip mount to separate from the friction clip through an opening in the friction clip.
20. The case of claim 14, wherein the removable kickstand comprises a plurality of friction clip mounts, and the case substrate comprises a plurality of corresponding friction clips removably coupled to one of the friction clip mounts.
